

SUMMARY

Kirchhoff's laws apply to voltage and current waveforms in the time domain and to the corresponding transforms in the s domain.

The s -domain models for the passive elements include initial condition sources and the element impedance or admittance. Impedance is the proportionality factor in the expression $V(s) = Z(s)I(s)$ relating the voltage and current transforms. Admittance is the reciprocal of impedance.

The impedances of the three passive elements are $Z_R = R$, $Z_L(s) = Ls$, and $Z_C(s) = 1/Cs$.

The s -domain circuit analysis techniques closely parallel the analysis methods developed for resistance circuits. Basic analysis techniques, such as circuit reduction, Thévenin's and Norton's theorems, the output method, or superposition, can be used on simple circuits. More complicated networks require a general approach, such as the node-voltage or mesh-current methods.

Response transforms are rational functions whose poles are zeros of the circuit determinant or poles of the transform of the input driving forces. Poles

introduced by the circuit determinant are called natural poles and lead to the natural response. Poles introduced by the input are called forced poles and lead to the forced response.

- In linear circuits, response transforms and waveforms can be separated into zero-state and zero-input components. The zero-state component is found by setting the initial capacitor voltages and inductor currents to zero. The zero-input component is found by setting all input driving forces to zero.
- The main purpose of s -domain circuit analysis is to gain insight into circuit performance without necessarily finding the time-domain response. The natural poles reveal the form, stability, and observability of the circuit's response. The number of natural poles is never greater than the number of energy storage elements in the circuit.
- A circuit is stable if all of its natural poles are in the left half of the s plane. Passive circuits are inherently stable so their natural poles are in the left half plane. Active circuits can be stable when circuit parameters are in one range and unstable for parameters outside this range.

PROBLEMS

PROBLEM 10-1 EQUIVALENT IMPEDANCE (Problems 10-1, 10-2)

In a linear circuit, use series and parallel equivalence to find the equivalent impedance at specified terminal pairs.

Problems 10-2, 10-4, 10-5, 10-6 and Exercises 10-3, 10-5

Find $Z_{EQ}(s)$ in Figure P10-1. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

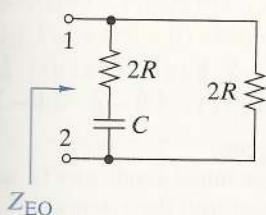


FIGURE P10-1

Find $Z_{EQ}(s)$ in Figure P10-2. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

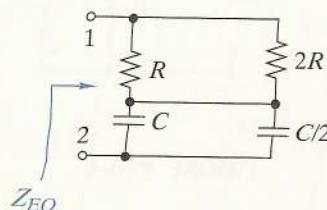


FIGURE P10-2

10-3 Find $Z_{EQ}(s)$ in Figure P10-3. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros for $R = 2 \text{ k}\Omega$, $L = 0.4 \text{ H}$, and $C = 100 \text{ nF}$.

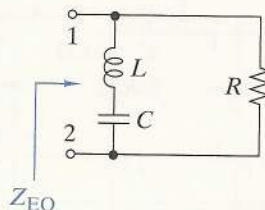


FIGURE P10-3

- 10-4 Find $Z_{EQ}(s)$ in Figure P10-4. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

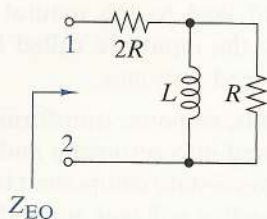


FIGURE P10-4

- 10-5 Find $Z_{EQ}(s)$ in Figure P10-5. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

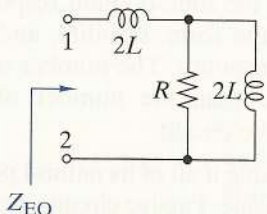


FIGURE P10-5

- 10-6 Find the poles and zeros of $Z_{EQ}(s)$ in Figure P10-6 with $R = 1 \text{ k}\Omega$, $L = 2 \text{ H}$, and $C = 500 \text{ nF}$.

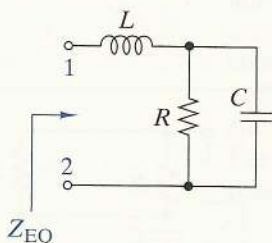


FIGURE P10-6

- 10-7 Find $Z_{EQ1}(s)$ and $Z_{EQ2}(s)$ in Figure P10-7. Express each impedance as a rational function and locate its poles and zeros.

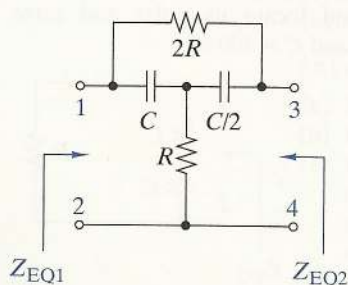


FIGURE P10-7

- 10-8 Find $Z_{EQ}(s)$ in Figure P10-8. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

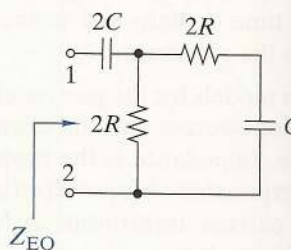


FIGURE P10-8

- 10-9 Find $Z_{EQ1}(s)$ and $Z_{EQ2}(s)$ in Figure P10-9. Express each impedance as a rational function and locate its poles and zeros.

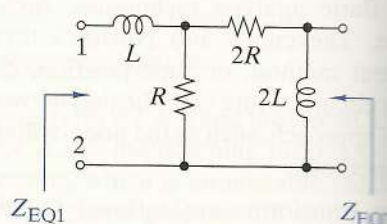


FIGURE P10-9

- 10-10 Find the equivalent impedance between terminals 1 and 2 in Figure P10-10. Select values of R and L such that $Z_{EQ}(s)$ has a pole at $s = -2000 \text{ rad/s}$. Locate the zeros of $Z_{EQ}(s)$ for your choice of R and L .

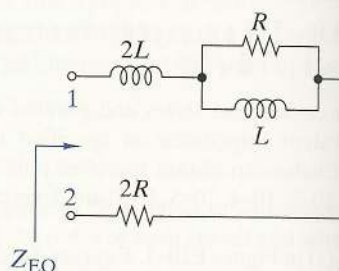


FIGURE P10-10

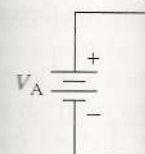
OBJECTIVE 10-2 BASIC CIRCUIT ANALYSIS TECHNIQUES (SECTS. 10-2, 10-3)

Given a linear circuit,

- Determine the initial conditions (if not given) and transform the circuit into the s domain.
- Solve for zero-state and/or zero-input response and waveforms using basic analysis methods and circuit reduction, unit output, Thévenin/Norton equivalent circuits, or superposition.

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Examples 10-1, 10-2, 1
Exercises 10-1, 1
10-10, 10-11

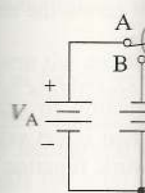
The switch in Figure 1
and is moved to
into the s domain
form.



FIGURE

The switch in Figure 1
and is moved to
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form.

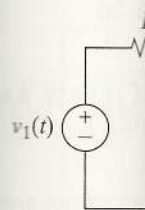
The switch in Figure 1
and is moved to
into the s domain
form.



FIGURE

The switch in Figure 1
and is moved to
into the s domain
form.

Transform the circuit
into the s domain
and find $i_L(s)$, $i_L(t)$, $V_L(s)$,
and $V_L(t)$.



FIGURE

locate the forced and natural poles, or select circuit parameters to place the natural poles at specified locations.

Examples 10-1, 10-2, 10-3, 10-4, 10-5, 10-6, 10-7, 10-8, and Exercises 10-1, 10-2, 10-3, 10-4, 10-5, 10-6, 10-7, 10-8, 10-9, 10-10, 10-11

The switch in Figure P10-11 has been in position A for a long time and is moved to position B at $t = 0$. Transform the circuit into the s domain and solve for $I_L(s)$ and $i_L(t)$ in symbolic form.

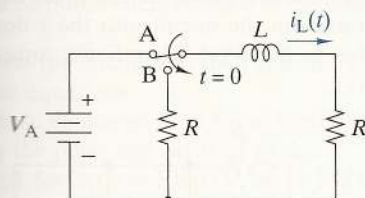


FIGURE P10-11

The switch in Figure P10-11 has been in position B for a long time and is moved to position A at $t = 0$. Transform the circuit into the s domain and solve for $I_L(s)$ and $i_L(t)$ in symbolic form.

The switch in Figure P10-13 has been in position A for a long time and is moved to position B at $t = 0$. Transform the circuit into the s domain and solve for $I_C(s)$ and $i_C(t)$ in symbolic form.

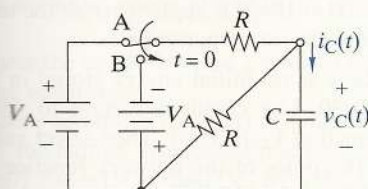


FIGURE P10-13

The switch in Figure P10-13 has been in position B for a long time and is moved to position A at $t = 0$. Transform the circuit into the s domain and solve for $V_C(s)$ and $v_C(t)$ in symbolic form.

Transform the circuit in Figure P10-15 into the s domain and find $I_L(s)$, $i_L(t)$, $V_L(s)$, and $v_L(t)$ when $v_1(t) = V_A u(t)$ and $i_L(0) = I_A$.

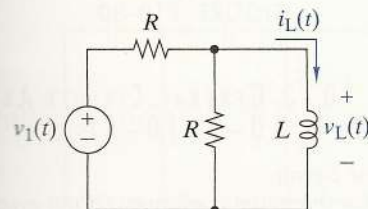


FIGURE P10-15

10-16 Transform the circuit in Figure P10-15 into the s domain and find $I_L(s)$, $i_L(t)$, $V_L(s)$, and $v_L(t)$ when $v_1(t) = V_A e^{-at} u(t)$ and $i_L(0) = I_A$.

10-17 The switch in Figure P10-17 has been in position A for a long time and is moved to position B at $t = 0$.

(a) Transform the circuit into the s domain and solve for $I_L(s)$ in symbolic form.

(b) Repeat part (a) using MATLAB.

(c) Find $i_L(t)$ for $R_1 = R_2 = 500 \Omega$, $R_3 = 1 \text{ k}\Omega$, $L = 250 \text{ mH}$, $C = 4 \mu\text{F}$, and $V_A = 15 \text{ V}$.

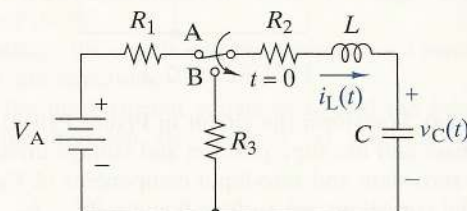


FIGURE P10-17

10-18 The switch in Figure P10-17 has been in position B for a long time and is moved to position A at $t = 0$.

(a) Transform the circuit into the s domain and solve for $V_C(s)$ in symbolic form.

(b) Find $v_C(t)$ for $R_1 = R_2 = 250 \Omega$, $R_3 = 1 \text{ k}\Omega$, $L = 500 \text{ mH}$, $C = 1 \mu\text{F}$, and $V_A = 15 \text{ V}$.

(c) Repeat part (b) using OrCAD.

10-19 The circuit in Figure P10-19 is in the zero state. The s -domain relationship between the input $I_1(s)$ and the output $I_R(s)$ is usually given as a ratio called a *network function*. Find $I_R(s)/I_1(s)$.

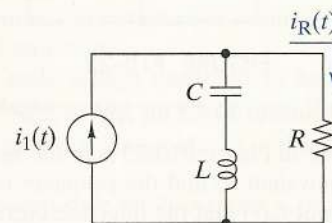


FIGURE P10-19

10-20 The circuit in Figure P10-20 is in the zero state. Find the s -domain relationship between the input $I_1(s)$ and the output $V_O(s)$.

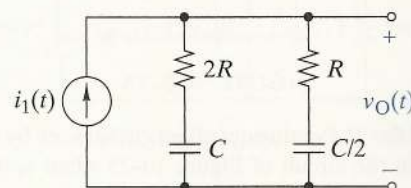


FIGURE P10-20

10-21 The initial conditions in Figure P10-21 are $v_C(0) = 0$ and $i_L(0) = I_0$. Transform the circuit into the s domain and use superposition and voltage division to find the zero-state and zero-input components of $V_C(s)$.

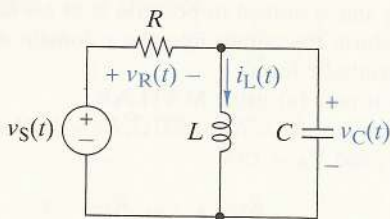


FIGURE P10-21

- 10-22 (a) Transform the circuit in Figure P10-21 into the s domain and use superposition and voltage division to find the zero-state and zero-input components of $V_R(s)$. The initial conditions are $v_C(0) = 0$ and $i_L(0) = I_0$.
 (b) Solve for $v_R(t)$ using OrCAD if $R = 1.5 \text{ k}\Omega$, $L = 500 \text{ mH}$, $C = 1 \text{ }\mu\text{F}$, $v_S(t) = 15u(t) \text{ V}$, $i_L(0) = 10 \text{ mA}$, and $v_C(0) = 0$.

10-23 There is no energy stored in the capacitor in Figure P10-23 at $t = 0$. Transform the circuit into the s domain and use current division to find $v_O(t)$ when the input is $i_S(t) = 2.5e^{-200t}u(t) \text{ mA}$. Identify the forced and natural poles in $V_O(s)$.

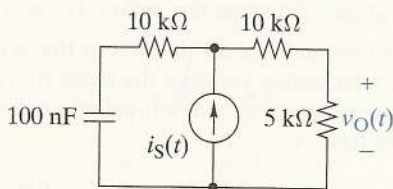


FIGURE P10-23

- 10-24 Repeat Problem 10-23 for $i_S(t) = 2.5e^{-400t}u(t) \text{ mA}$.
 10-25 The circuit in Figure P10-25 is in the zero state. Use a Thévenin equivalent to find the s -domain relationship between the input $I_S(s)$ and the interface current $I(s)$.

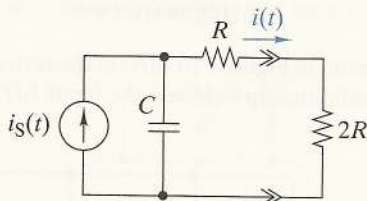


FIGURE P10-25

- 10-26 Find the Thévenin equivalent circuit seen by the $2R$ load resistor in the circuit of Figure 10-25 when $v_C(0) = V_0$.
 10-27 The circuit in Figure P10-27 is in the zero state. Find the Thévenin equivalent to the left of the interface.

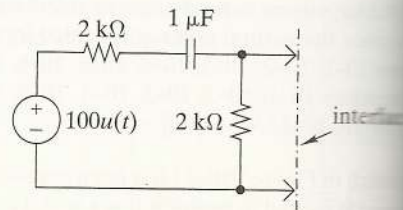


FIGURE P10-27

10-28 There is no initial energy stored in the circuit in Figure P10-28. Transform the circuit into the s domain and use superposition to find $V(s)$. Identify the forced and natural poles in $V(s)$.

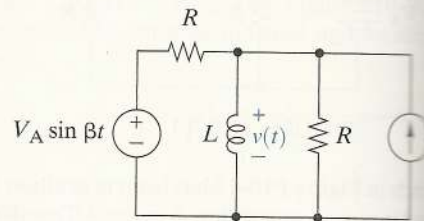


FIGURE P10-28

10-29 The equivalent impedance between a pair of terminals is

$$Z_{EQ}(s) = 1000 \frac{s + 2000}{s + 1000}$$

A voltage $v(t) = 10u(t)$ is applied across the terminals. Find the resulting current response $i(t)$.

10-30 There is no initial energy stored in the circuit in Figure P10-30. Use circuit reduction to find the network function $V_2(s)/V_1(s)$. Then select values for R and C so that the poles of the network function are approximately -2618 and -382 rad/s .

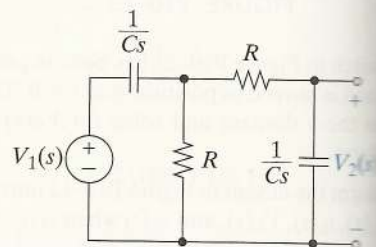


FIGURE P10-30

OBJECTIVE 10-3 GENERAL CIRCUIT ANALYSIS (SECTS. 10-4, 10-5, 10-6)

Given a linear circuit:

- (a) Determine the initial conditions (if not given) and transform the circuit into the s domain.

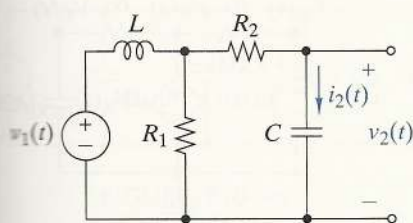


FIGURE P10-31

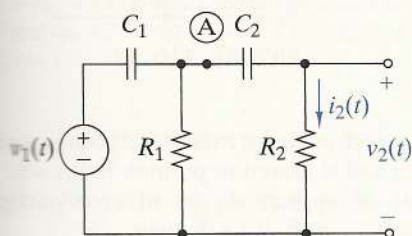


FIGURE P10-33

10-34 There is no initial energy stored in the circuit in Figure P10-33. The Thévenin equivalent circuit to the left of point A when a unit step is applied is

$$V_T(s) = \frac{1}{s + 10^3} \text{ V-s, and } Z_T(s) = \frac{10^6}{s + 10^3} \Omega$$

Select values for R_2 and C_2 such that the output transform is

$$V_2(s) = \frac{s}{s^2 + 3000s + 10^6} \text{ V-s}$$

10-35 There is no initial energy stored in the bridged-T circuit in Figure P10-35.

(a) Transform the circuit into the s domain and formulate mesh-current equations.

(b) Use the mesh-current equations to find the s -domain relationship between the input $V_1(s)$ and the output $V_2(s)$.

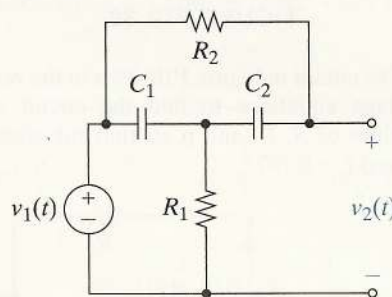


FIGURE P10-35

10-36 There is no initial energy stored in the bridged-T circuit in Figure P10-35.

(a) Transform the circuit into the s domain and formulate node-voltage equations.

(b) Use the node-voltage equations to find the s -domain relationship between the input $V_1(s)$ and the output $V_2(s)$.

10-37 There is no initial energy stored in the circuit in Figure P10-37.

(a) Find the zero-state mesh currents $i_A(t)$ and $i_B(t)$ when $v_1(t) = 100e^{-200t}u(t)$ V.

(b) Validate your answers using OrCAD.

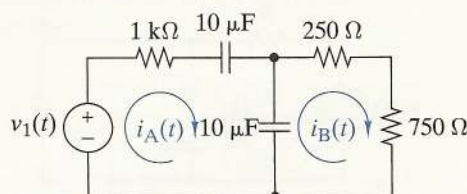


FIGURE P10-37

10-38 There is no external input in the circuit in Figure P10-38.